



Slides 1.6: Measuring and the Silent Measurement Challenge

Lesson 1.6, Unit 1, Topic 1.2. Two days. Day 1 is slides 1 to 11; Day 2 is slides 12 to 22.

Day 1

Slide 1: Do now

- A recipe says 1 c flour.
- Your friend scoops the cup into the bag and packs it down.
- Is that 1 cup? Yes or no, and why. Two minutes.

Slide 2: The scale does not lie

- One cup of flour, packed: about 150 grams.
- One cup of flour, spooned and leveled: about 120 grams.
- That is a quarter cup extra. That is why the cookies were bricks. Image: two measuring cups on a scale, side by side, with the two readings. Notes: Do this live. Pack one, spoon the other, weigh both. Let them see the number.

Slide 3: Three kinds, three tools

- Dry (flour, sugar, cocoa): dry measuring cups.
- Liquid (water, milk, oil): liquid measuring cup.
- Small amounts (salt, baking powder, vanilla): measuring spoons. Image: the three tools with the three ingredient groups.

Slide 4: Dry technique

- Spoon it into the cup. Do not scoop, do not pack, do not tap.
- Fill over the top.
- Level off with a straight edge (the back of a butter knife).
- Exception: brown sugar is packed. The recipe will say so. Image: a hand leveling a cup of flour with a butter knife, flour falling into a tray.

Slide 5: Liquid technique

- Set the glass cup on the counter. Do not hold it in the air.
- Pour to the line.
- Bend down. Read at eye level. The bottom of the curve (the meniscus) sits on the line.
- From above, you will read it high every time. Image: a side view of a liquid measuring cup at eye level with the meniscus on the 1 cup line.

Slide 6: Small amounts

- Dry: fill the spoon, level it.
- Liquid: fill to the brim.
- Do it over a bowl, not over the mix. A spill goes in the bowl, not the recipe.
- 1/4 tsp is real. It matters. Baking powder proves it. Image: measuring spoons fanned out with the sizes labeled.

Slide 7: Abbreviations

- tsp = teaspoon

- Tbsp = tablespoon (the big T is the big spoon)
- c = cup
- oz = ounce (fl oz for liquids)
- lb = pound Image: each abbreviation next to a picture of the tool.

Slide 8: Equivalents (say them twice)

- 3 tsp = 1 Tbsp
- 16 Tbsp = 1 c
- 4 Tbsp = $\frac{1}{4}$ c
- 8 fl oz = 1 c (liquid)
- 2 c = 1 pint; 16 oz = 1 lb (by weight) Notes: Chant them. Then cover the right side and chant again.

Slide 9: Practice stations

- A flour, B sugar, C water, D oil, E salt.
- Two amounts at each station. Right tool, right technique.
- If your station has a scale, weigh the first amount and write the grams.
- Return the ingredient to the container. Spills go on the tray.
- Three minutes per station.

Slide 10: Equivalents practice

- Six problems on the handout, Part 3.
- Alone first. Then compare with a partner.
- Problem 6 is a real one: rice for 4 people becomes rice for 8.
- Grade 8: problems 7 and 8.

Slide 11: Closure

- Exit card: 3 equivalents from memory, 2 techniques you will use tomorrow, 1 thing that still confuses you.
- Tomorrow: nobody talks. Six measurements. One scale. Points.

Day 2

Slide 12: Do now

- Write one way to tell a teammate “stop, that is too much” without using your voice.
- Two minutes. Then two people show theirs.

Slide 13: The Silent Measurement Challenge

- Six measurements on your team card.
- One scale. The scale is the judge.
- Points for accuracy. 18 possible.
- Nobody talks. Image: a tray with six numbered cups and a scale beside it.

Slide 14: The rules

- No talking, no notes, no mouthing words.
- Pointing, gestures, fingers, and showing the card are allowed.
- A different person measures each cup. Everyone else watches and signals.
- Once a cup is on the tray, it is done. Do not touch it again.
- Right tool for each kind. Carry the tray to the weigh-in together. Notes: Model one silent correction: hold up a flat hand, point at the line on the cup.

Slide 15: Points

- 3: in the tight band.
- 2: in the wide band.
- 1: right ingredient, wrong amount.
- 0: wrong ingredient or empty cup.
- Team totals go on the board.

Slide 16: Go

- 12 minutes.
- Time calls at 6 and 10.
- I say nothing else. Image: a large timer.

Slide 17: Weigh-in

- Tare the scale on an empty cup.
- Each cup, one at a time. I read the grams out loud.
- Record your grams and points on the handout, Part 4.
- Mark the cup that missed. Image: the scoreboard grid with four team rows.

Slide 18: Why measuring matters, baking chemistry

- Flour is the structure.
- Sugar is the sweetness and the browning.
- Baking powder is the lift.

- Fat (oil, butter) is the tenderness.
- Too much of one throws off the others. A cake is a ratio. Image: a simple bar for each ingredient labeled with its job.

Slide 19: Reading the miss

- Flour heavy: packed or scooped.
- Flour light: liquid cup read from above, or tapped and settled.
- Sugar off: not leveled.
- Water off: read from above (heavy) or spilled on the way (light).
- Salt heavy: a heaped spoon.
- Which one was your team's miss?

Slide 20: How did you talk?

- You could not use words. What did you use?
- The signal that worked.
- The moment you wished you could talk.
- This has a name: nonverbal communication. Cooks use it in every loud kitchen. Notes: One team at a time. Write the signals on the board. Say out loud that the students who are quiet in English are often the best at this game.

Slide 21: Grade 8 stretch

- Percent error for each cup: (your grams minus target) divided by target, times 100.
- Which cup had the biggest percent error? Was it the biggest gram error?

Slide 22: Closure

- Exit card, one sentence: "Next time I measure ___ I will ___ because the scale showed ___."
- Two students read theirs.
- Tomorrow: reading the recipe you will cook.